Section 3.2 Matrix Algebra

Matrices are mathematical objects and we can do fun things like add them and multiply them. The set of $n \times n$ matricies itself makes up something called a non-communitave Ring, not a topic for this class but worth a google if you are intrested. A Ring, in short, is an algebraic structure for which it's members satisify specific properties.

So we know how to add vectors but how we add matrices? If $A, B \in R^{n \times m}$, the set of $n \times m$ matrices.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \qquad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix}$$

Then

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1m} + b_{1m} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2m} + b_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & a_{n2} + b_{n2} & \cdots & a_{nn} + b_{nm} \end{bmatrix}$$

That is we add them component wise.

Example: Calculate $A + B = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 2 & 7 & 4 \\ 0 & 8 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 4 & 3 \\ 6 & 1 & 0 \\ 2 & 5 & 2 \\ 9 & 1 & 10 \end{bmatrix}$

What about $B + A$?

Section 3.2 Matrix Algebra

Scalar Multiplication

Let $c \in R$ and $A \in R^{n\times m}$ be a scalar then

$$cA = cA = c\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1m} \\ ca_{21} & ca_{22} & \cdots & ca_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{nm} \end{bmatrix}$$

Example: Calculate

$$4\begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 2 & 7 & 4 \\ 0 & 8 & 1 \end{bmatrix}$$

Theorem 3.11: Let s and t be scalars and $A, B, C \in R^{n\times m}$ and 0_{nm} a matrix of zeros of size $n \times m$ then,

$(a)\ A + B = B + A$
$(b)\ s(A + B) = sA + sB$
$(c)\ (s + t)A = sA + tA$
$(d)\ (A + B) + C = A + (B + C)$
$(e)\ (st)A = s(tA)$
$(f)\ (A + 0_{nm}) = A$

Section 3.2 Matrix Algebra

Matrix Multiplication

Let $A = [\vec{a}_1 \;\; \vec{a}_2 \;\; \cdots \;\; \vec{a}_k] \in R^{n\times k}$ whre $\vec{a}_i \in R^n$ and $B = [\,\vec{b}_1 \;\; \vec{b}_2 \;\; \cdots \;\; \vec{b}_m] \in R^{k\times m}$ and $\vec{b}_i \in R^k$ then the matrix product is defined by:

$$AB = \left[A\vec{b}_1 \;\; A\vec{b}_2 \;\; \cdots \;\; A\vec{b}_m\right]$$

$$= \left[(b_{11}\vec{a}_1 + b_{21}\vec{a}_2 + \cdots + b_{k1}\vec{a}_k) \;+\; (b_{12}\vec{a}_1 + b_{22}\vec{a}_2 + \cdots + b_{k2}\vec{a}_k) + \cdots + (b_{1m}\vec{a}_1 + b_{2m}\vec{a}_2 + \cdots + b_{km}\vec{a}_k)\right]$$

$$= \left[\left(b_{11}\begin{bmatrix}a_{11}\\a_{21}\\\vdots\\a_{n1}\end{bmatrix} + b_{21}\begin{bmatrix}a_{12}\\a_{22}\\\vdots\\a_{n2}\end{bmatrix} + \cdots + b_{k1}\begin{bmatrix}a_{1k}\\a_{2k}\\\vdots\\a_{nk}\end{bmatrix}\right) + \left(b_{12}\begin{bmatrix}a_{11}\\a_{21}\\\vdots\\a_{n1}\end{bmatrix} + b_{22}\begin{bmatrix}a_{12}\\a_{22}\\\vdots\\a_{n2}\end{bmatrix} + \cdots + b_{k2}\begin{bmatrix}a_{1k}\\a_{2k}\\\vdots\\a_{nk}\end{bmatrix}\right) + \cdots + \left(b_{1m}\begin{bmatrix}a_{11}\\a_{21}\\\vdots\\a_{n1}\end{bmatrix} + b_{2m}\begin{bmatrix}a_{12}\\a_{22}\\\vdots\\a_{n2}\end{bmatrix} + \cdots + b_{km}\begin{bmatrix}a_{1k}\\a_{2k}\\\vdots\\a_{nk}\end{bmatrix}\right)\right]$$

$$= \left[\begin{bmatrix}b_{11}a_{11} + b_{21}a_{12} + \cdots + b_{k1}a_{1k}\\b_{11}a_{21} + b_{21}a_{22} + \cdots + b_{k1}a_{2k}\\\vdots\\b_{11}a_{n1} + b_{21}a_{n2} + \cdots + b_{k1}a_{nk}\end{bmatrix} \;\; \begin{bmatrix}b_{12}a_{11} + b_{22}a_{12} + \cdots + b_{k2}a_{1k}\\b_{12}a_{21} + b_{22}a_{22} + \cdots + b_{k2}a_{2k}\\\vdots\\b_{12}a_{n1} + b_{22}a_{n2} + \cdots + b_{k2}a_{nk}\end{bmatrix} \;\; \cdots \;\; \begin{bmatrix}b_{1m}a_{11} + b_{2m}a_{12} + \cdots + b_{km}a_{1k}\\b_{1m}a_{21} + b_{2m}a_{22} + \cdots + b_{km}a_{2k}\\\vdots\\b_{1m}a_{n1} + b_{2m}a_{n2} + \cdots + b_{km}a_{nk}\end{bmatrix}\right]$$

We can also state this in the following way if $AB = C$ then

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ik}b_{kj}$$

Which is an $n \times m$ matrix

Example: Calculate $AB = \begin{bmatrix}1 & 0\\-1 & 2\\3 & 1\end{bmatrix}\begin{bmatrix}2 & -2 & 1 & 1\\4 & 2 & -1 & 0\end{bmatrix}$

One way to think about it is that the entry in the ith row and j^{th} column will come from the ith row of the left matrix and the jth column of the right matrix.

What about BA?

Section 3.2 Matrix Algebra

In General $AB \neq BA$ for matricies!

Matrices of different sizes

```
octave:2> A=[1,2,3,4;5,6,7,8]
A =

   1   2   3   4
   5   6   7   8

octave:3> B=[1 2; 3 4; 5 6; 7 8]
B =

   1   2
   3   4
   5   6
   7   8

octave:4> A*B
ans =

    50    60
   114   140

octave:5> B*A
ans =

   11   14   17   20
   23   30   37   44
   35   46   57   68
   47   62   77   92
```

Matrices of the same size

```
octave:6> A=[1,2,3;4,5,6;7,8,9]
A =

   1   2   3
   4   5   6
   7   8   9

octave:7> B=[9,8,7;6,5,4;3,2,1]
B =

   9   8   7
   6   5   4
   3   2   1

octave:8> A*B
ans =

    30    24    18
    84    69    54
   138   114    90

octave:9> B*A
ans =

    90   114   138
    54    69    84
    18    24    30
```

Notice here they were the same, but this was **not** matrix multiplication. In Matlab or Octave * and .* (dot times) are very different. The .* operator multiplies component wise $a_{11}b_{11}$ $a_{12}b_{12}$ etc. In fact it is the Hadamard product we mentioned when first talking about vectors. If the matricies are of different sides you cannot do the Hadamard product.

```
octave:10> A.*B
ans =

    9   16   21
   24   25   24
   21   16    9

octave:11> B.*A
ans =

    9   16   21
   24   25   24
   21   16    9
```

Section 3.2 Matrix Algebra

We stated in Theorem 3.11 that $A + 0_{nm} = A$ the zero matrix is often refered to as the additive identity for matrix addition, one might wonder then, is there a mltiplicative identity for matrix multiplication?

The identity matrix: Let $A \in R^{n\times m}$ then the identity matricies will be $I_m \in R^{m\times m}$, $I_n = R^{n\times n}$ and will have ones on the diagonal and zeros everywhere else. That is:

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \vdots \\ \vdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$ Further $AI_m = I_n A = A$

If $A \in R^{n\times n}$, is a square matrix $n = m$, then $I = \ I_n = I_m \in R^{n\times n}$ and we can say $AI = IA = A$

Example:

Multiply:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Section 3.2 Matrix Algebra

Theorem 3.13: Let s be a scalar and A, B, C be matrices of appropriate sizes. Then

$(a)\ A(BC) = (AB)C$
$(b)\ A(B + C) = AB + AC$
$(c)\ (A + B)C = AC + BC$
$(d)\ s(AB) = (sA)B = A(sB)$
$(e)\ AI = A$
$(f)\ IA = A$

Theorem 3.14 Let A, B, C be nonzero matrices then,
(a) In general $AB \neq BA$
(b) $AB = 0$ **does not** imply that $A = 0$ or $B = 0$.
(c) $AC = BC$ **does not** imply that $A = B$ or $C = 0$

Example $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}\begin{bmatrix} -4 & 2 \\ 2 & -3 \end{bmatrix}$

Example Let $A = \begin{bmatrix} -3 & 3 \\ 11 & -3 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ Find AB *and* BC

Section 3.2 Matrix Algebra

The Transpose of a Matrix

The transpose of a matrix is easy, columns become rows and rows become rows become columns

The first column becomes the first row, the second column becomes the second row, the third column becomes the third row and so on.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{\text{Transpose...}} A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Notice the diagonal stayed the same! Infact we are just moving $a_{ij}\ \ to\ a_{ji}$ so on the diagonal $a_{ii} = a_{ii}$

Theorem 3.15: Let $A, B \in R^{n\times m}, C \in R^{m\times k}$ and s scalar. Then

$(a)\ \ (A+B)^T = A^T + B^T$
$(b)\ \ (sA)^T = sA^T$
$(c)\ \ (AC)^T = C^T A^T$

Quiz problem: Prove (a)

Section 3.2 Matrix Algebra

Powers of a matrix

Just like we would say x multiplied by itself n times is written x^n we can do the same with a matrix

$$A^n = AA\cdots A \text{ where } A \text{ is multiplied by itself } n \text{ times}$$

Diagonal matrices are particularly nice for this operation.

$$A^n = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}\cdots\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} a_{11}^n & 0 & 0 & \cdots & 0 \\ 0 & a_{22}^n & 0 & \cdots & 0 \\ 0 & 0 & a_{33}^n & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn}^n \end{bmatrix}$$

Ex: Calculate A^3 if $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Another important form of matrices are Upper Triangular and Lower Triangular matrices.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Upper Triangular

$$\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

Lower Triangular

Note that an Upper triangular matrix corresponds to echelon form and a diagonal matrix is almost in reduced row echelon form. We can also make the following statements.

(a) *Upper and Lower Triangular matrices are called triangular*
(b) *A Diagonal matrix is both Upper Triangular and Lower Triangular*
(c) *If A and B are both either Upper Triangular or Lower Triangular so is* $A + B$
(d) *Powers of triangular matricies are also triangular*

Section 3.2 Matrix Algebra

Elementary Matrices

We have been doing row operations on matrices of late, it turns out that we can represent those row operations with matrix multiplication using Elementary Matrices.

We form an elementary matrix by doing a **single** desired row operation on the Identity matrix. After that row operation is done we can use the resulting matrix, call it E, and multiply it by a matrix A . The result of the multiplication will be the same as doing the row operation on A

Example: Let $A = \begin{bmatrix} -2 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix}$ find a matrix E_1 such that E_1A produces the same result as swapping R_1, R_2 and a matrix E_2 such that E_2A produces the result of multiplying Row 1 by 2 and adding it to Row 1, that is $2R_1 + R_2$

What is the effective result of E_2E_1A?

Section 3.2 Matrix Algebra

Composition of Linear Transformations.

In the last section we discussed how linear transformations on vectors in R^n can be represented by matrix multiplication. That is for any linear trasformation $T_1(\vec{u})$ there is some matrix A_1 such that

$$T_1(\vec{u}) = A_1\vec{u}$$

Imagine now that we have more than one linear transformation which we wish to compose

Let $T_1: R^n \to R^m$ and $T_2: R^m \to R^k$ The composotion of linear transformations can be written

$$W(\vec{u}) = T_2(T_1(\vec{u})): R^n \to R^k$$

Suppose $T_1(\vec{u}) = A_1\vec{u}$ and $T_2(\vec{v}) = A_2\vec{v}$ Then $W(\vec{u}) = B\vec{u}$ Where $B = A_2A_1$ and $B \in R^{k\times n}$

Example

Let $T_1\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_1 \\ x_1 - x_3 \\ 2x_1 + x_2 \end{bmatrix}$ $T_2\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 - x_3 \\ 2x_1 - x_2 - x_3 \end{bmatrix}$ Find A_1 and A_2

Find A_2A_1

Section 3.2 Matrix Algebra

$T_1\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_1 \\ x_1 - x_3 \\ 2x_1 + x_2 \end{bmatrix}$ $T_2\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 - x_3 \\ 2x_1 - x_2 - x_3 \end{bmatrix}$

Calculate $T_2\left(T_1\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right)\right)$

Verify $A_2A_1\vec{x}$ gives the same result for $\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

Section 3.2 Matrix Algebra

Partitioned Matrix

When we think about the size of a matrix $n \times m$ we can calcualte how many numbers are in the matrix. nm . In many applied settings macticies can have tens of thousands or more rows and columns. So if a matrix has 10,000 rows and 10,000 columns you have 10^6 numbers. Then we want to add multiply and divide those numbers in computation, this can really eat up computer memory and CPU capacity. To aid in this we can split up our matricies into smaller matricies and carry out what ever computations a piece at a time to save RAM if needed or to parallelize the problem by working on several blocks at once.

$$A = \left[\begin{array}{cc|ccc} 1 & -2 & 3 & 0 & 1 \\ 2 & 0 & 1 & 2 & 3 \\ -1 & 4 & 5 & 4 & 1 \\ \hline 4 & 1 & 2 & 2 & -3 \end{array}\right] \qquad B = \left[\begin{array}{cc|ccc} 0 & 1 & 4 & 0 & 2 \\ 2 & 1 & 2 & 4 & 4 \\ -1 & 3 & 5 & 6 & -1 \\ \hline -2 & 3 & 1 & 0 & 2 \end{array}\right]$$

Here we have partitioned the matrices into what we call Block Matrices

$$A_{11} = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ -1 & 4 \end{bmatrix} \qquad A_{12} = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 2 & 3 \\ 5 & 4 & 1 \end{bmatrix} \qquad B_{11} = \begin{bmatrix} 0 & 1 \\ 2 & 1 \\ -1 & 3 \end{bmatrix} \qquad B_{12} = \begin{bmatrix} 4 & 0 & 2 \\ 2 & 4 & 4 \\ 5 & 6 & 1 \end{bmatrix}$$

$$A_{21} = [4 \quad 1] \qquad A_{22} = [2 \quad 2 \quad -3] \qquad B_{21} = [-2 \quad 3] \qquad B_{22} = [1 \quad 0 \quad 2]$$

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

We can then represent our original matrices with the block matrices.

$$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

Addition

$$A + B = \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} \\ A_{21} + B_{21} & A_{22} + B_{22} \end{bmatrix}$$

Section 3.2 Matrix Algebra

Multiplying Partitioned Matrices

When adding partitioned matrices the matrices needed to be the same size and partitioned in the same way. However, when we multiply matrices they don't necessarily need to be the same size. In addition we have to make sure we partition them in a way that the block matrices themselves can be multiplied. That is so that their inner dimensions agree.

Example.

$$A = \left[\begin{array}{cc|ccc} 1 & -2 & 3 & 0 & 1 \\ 2 & 0 & 1 & 2 & 3 \\ -1 & 4 & 5 & 4 & 1 \\ \hline 4 & 1 & 2 & 2 & -3 \end{array}\right] \qquad B = \left[\begin{array}{c|c|c} 1 & 0 & 4 \\ -1 & 1 & 2 \\ \hline 5 & 2 & 2 \\ -2 & 1 & 0 \\ 1 & 1 & 2 \end{array}\right]$$

$$A_{11} = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ -1 & 4 \end{bmatrix} \qquad A_{12} = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 2 & 3 \\ 5 & 4 & 1 \end{bmatrix} \qquad B_{11} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad B_{12} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad B_{13} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$A_{21} = [4 \quad 1] \qquad A_{22} = [2 \quad 2 \quad -3] \qquad B_{21} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} \quad B_{22} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad B_{23} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \qquad B = \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \end{bmatrix}$$

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} & A_{11}B_{13} + A_{12}B_{23} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} & A_{21}B_{13} + A_{22}B_{23} \end{bmatrix}$$